Scenario for Project 2

Heaven Classics successfully creates an EC2 Server Instance for Windows 2012 Server. After launching the instance on the server, the next step was to monitor the operations.

Monitoring is important to keep an eye on the performance of an EC2 instance. It helps gather data from all parts, and is useful for debugging failure.

The monitoring team at Heaven Classics started monitoring activities using the CloudWatch Service in the AWS Management Console. The Heaven Classics support team were required to meet the following objectives:

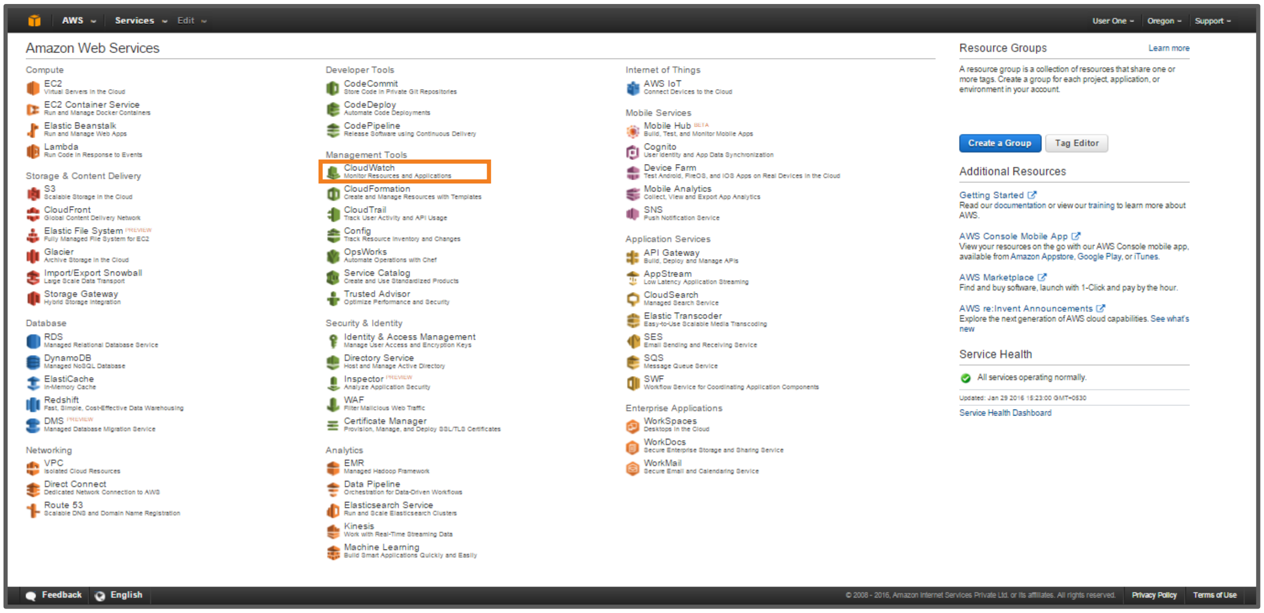
1. Check the CPU Utilization.
2. Create an Alarm.
3. Create an IAM User.
4. Create the IAM Administrator Group, and add the user to the Administrator Group.
5. Create a Role.

1. Check the CPU Utilization

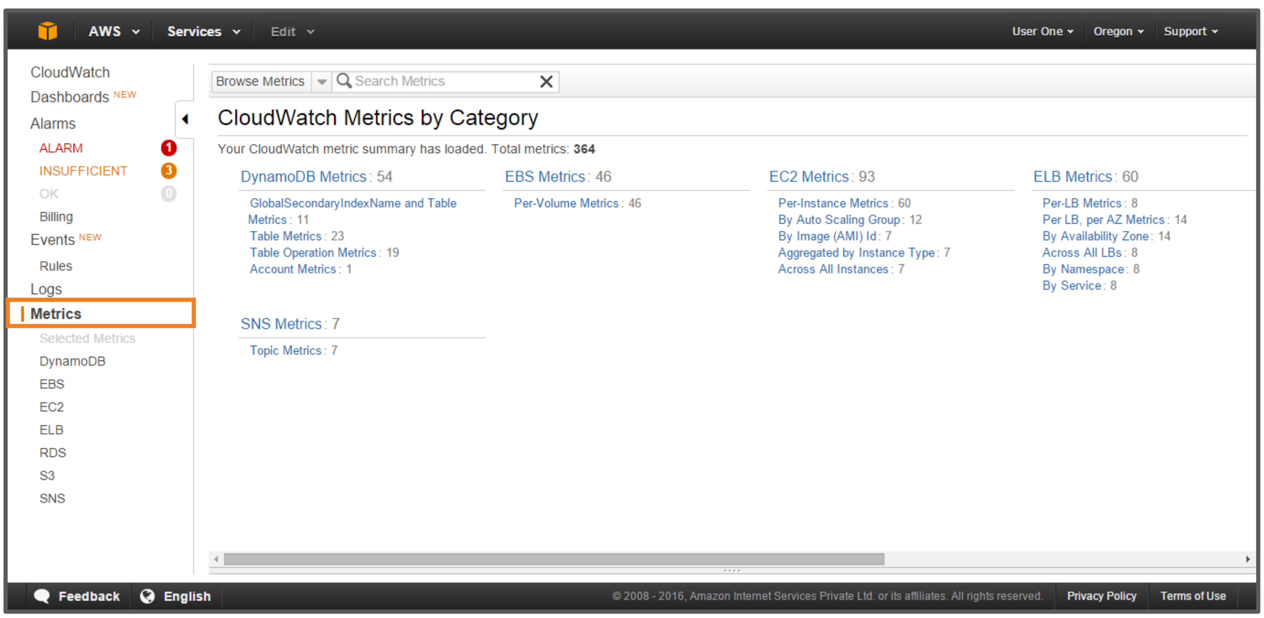
After deciding the monitoring goals, the next step is to decide the minimum usage for an EC2 instance, and monitoring it.

To check the CPU Utilization:

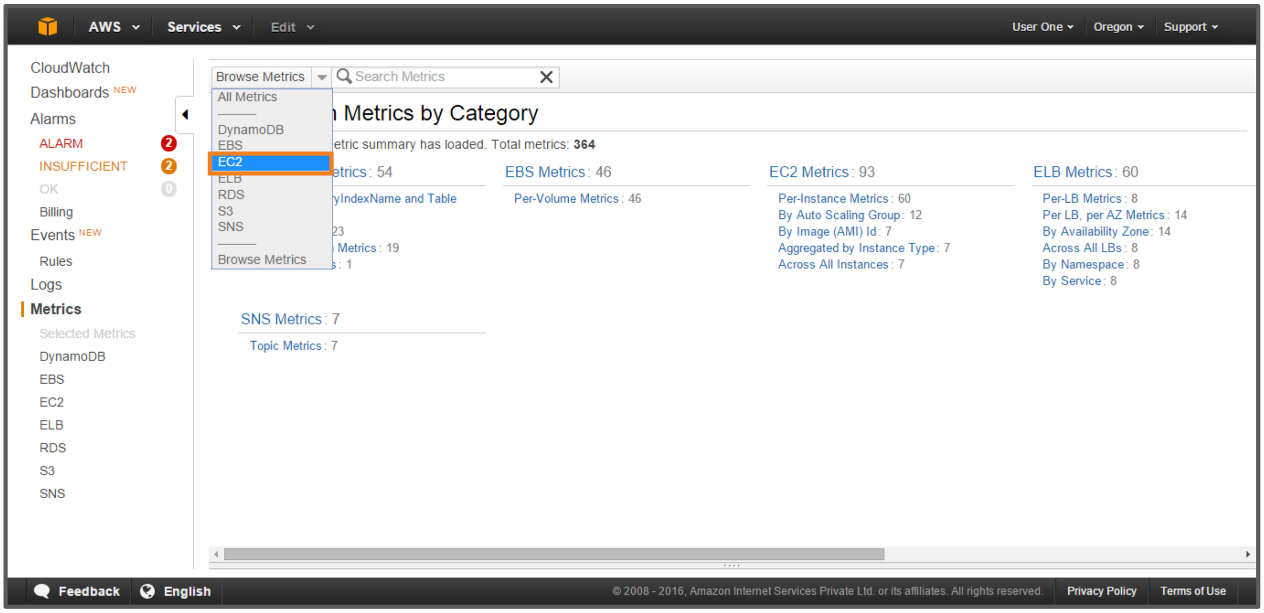
1. Sign in to the AWS Management Console to open the CloudWatch Console.



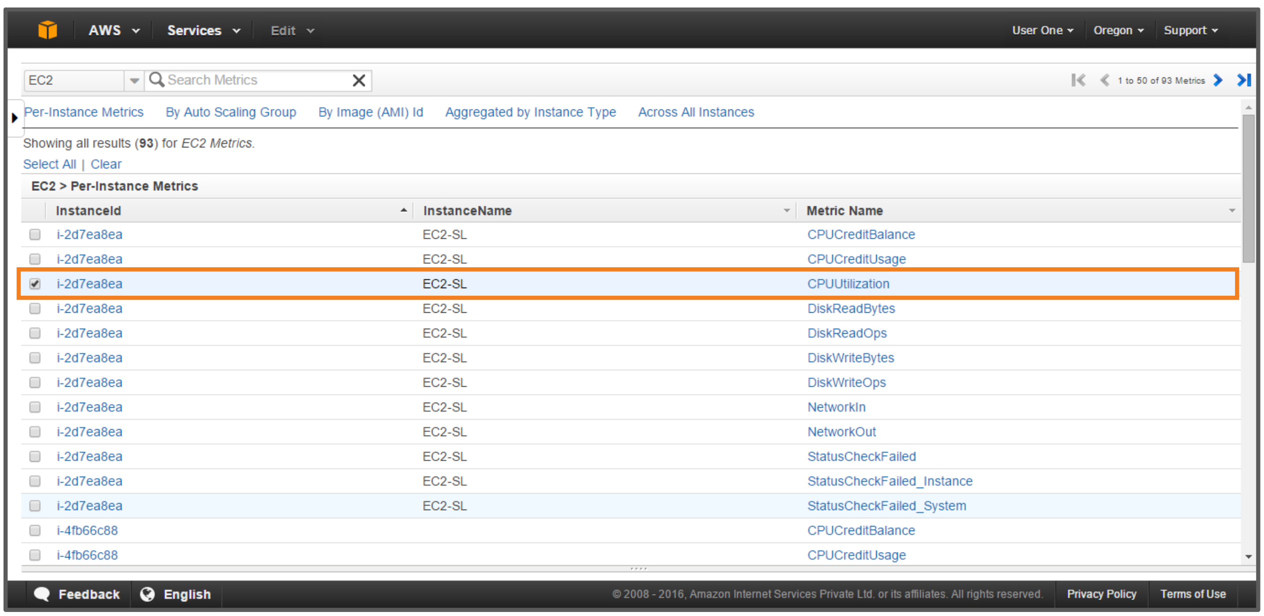
2. On the Navigation pane, click **Metrics**.



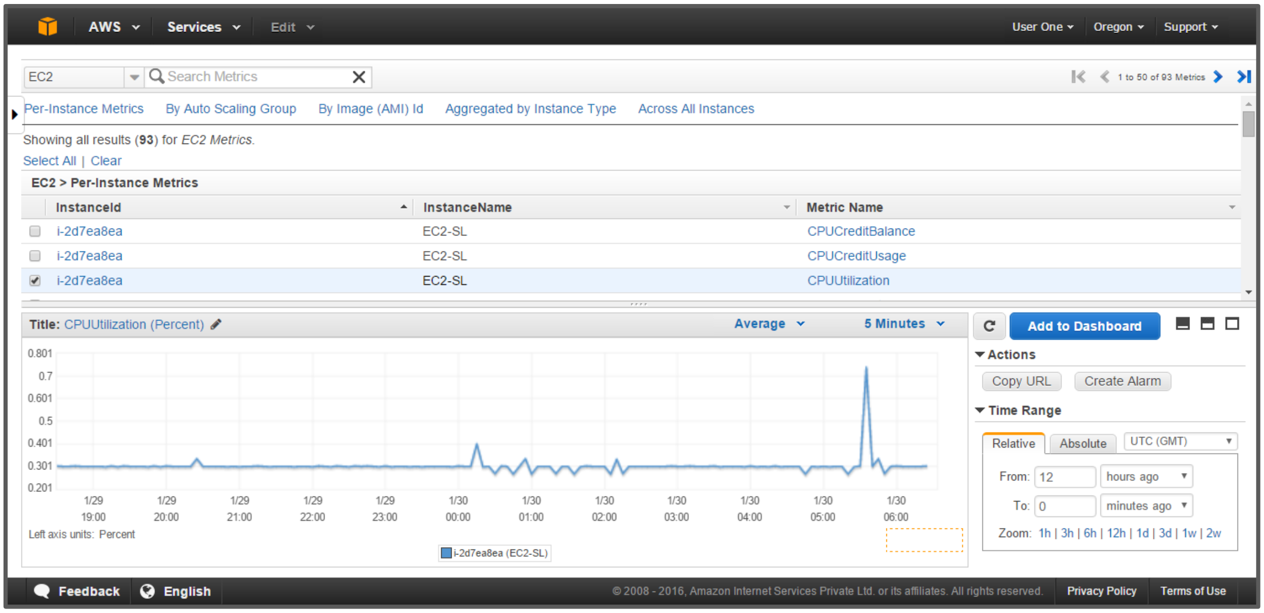
3. Then, in the **CloudWatch Metrics by Category** pane, select the **EC2** Metrics category from the Browse Metrics drop down list.



4. Next, select a row of metric that contains **CPUUtilization**.



5. After selecting CPUUtilization, look for the graph that presents average CPUUtilization for a single instance in the details pane. This shows the CPU utilization. The monitoring team can review this graph while monitoring the system resources.


AWS
Services
Edit
User One
Oregon
Support
EC2
Search Metrics
1 to 50 of 93 Metrics
Per-Instance Metrics
By Auto Scaling Group
By Image (AMI) Id
Aggregated by Instance Type
Across All Instances
Showing all results (93) for EC2 Metrics.
Select All | Clear
EC2 > Per-Instance Metrics
InstanceId
InstanceName
Metric Name
i-2d7ea8ea
EC2-SL
CPUCreditBalance
i-2d7ea8ea
EC2-SL
CPUCreditUsage
i-2d7ea8ea
EC2-SL
CPUUtilization
Title: CPUUtilization (Percent)
Average
5 Minutes
Add to Dashboard
Actions
Copy URL
Create Alarm
Time Range
Relative
Absolute
UTC (GMT)
From: 12 hours ago
To: 0 minutes ago
Zoom: 1h | 3h | 6h | 12h | 1d | 3d | 1w | 2w
Left axis units: Percent
i-2d7ea8ea (EC2-SL)
Feedback
English
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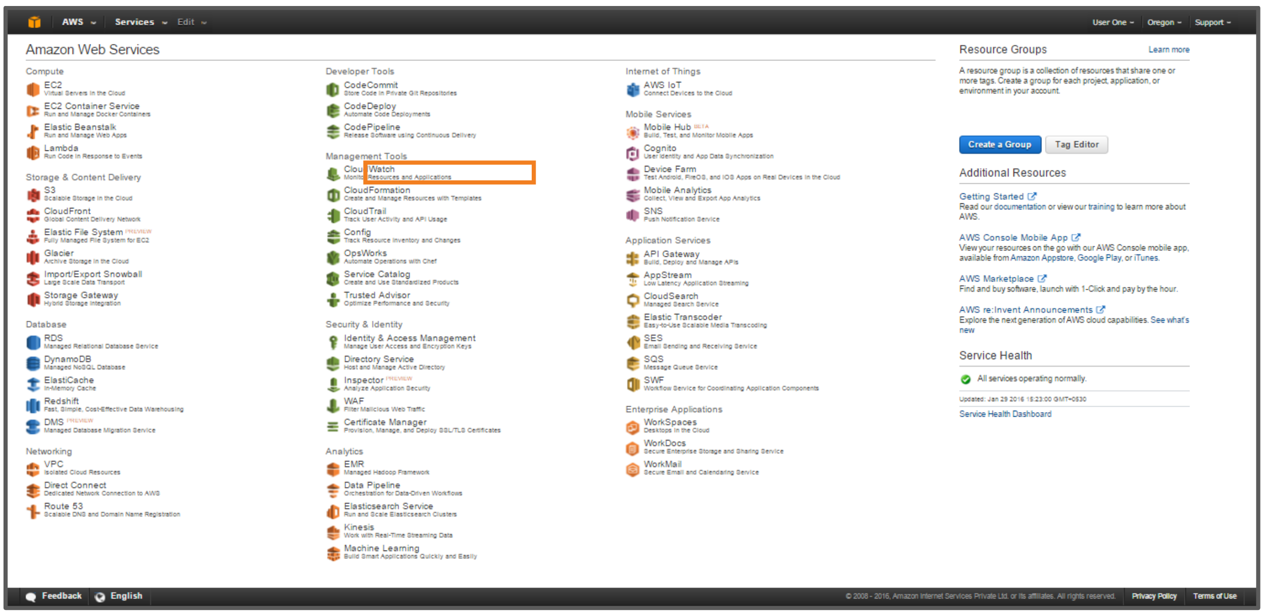
2. Create an Alarm

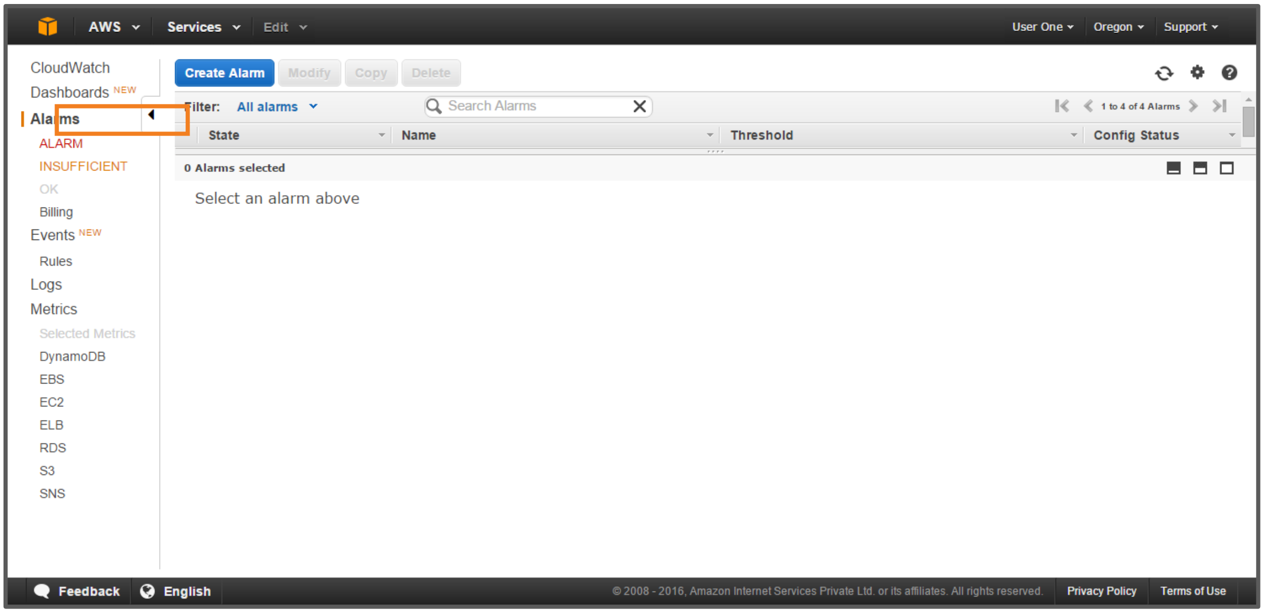
Along with monitoring the CPU utilization, the support team sets an alarm to send notifications to the monitoring team. This helps them in quick tracking the current state of the system, and taking appropriate action if needed. Let's now see the steps to create an alarm.

To create an Alarm:

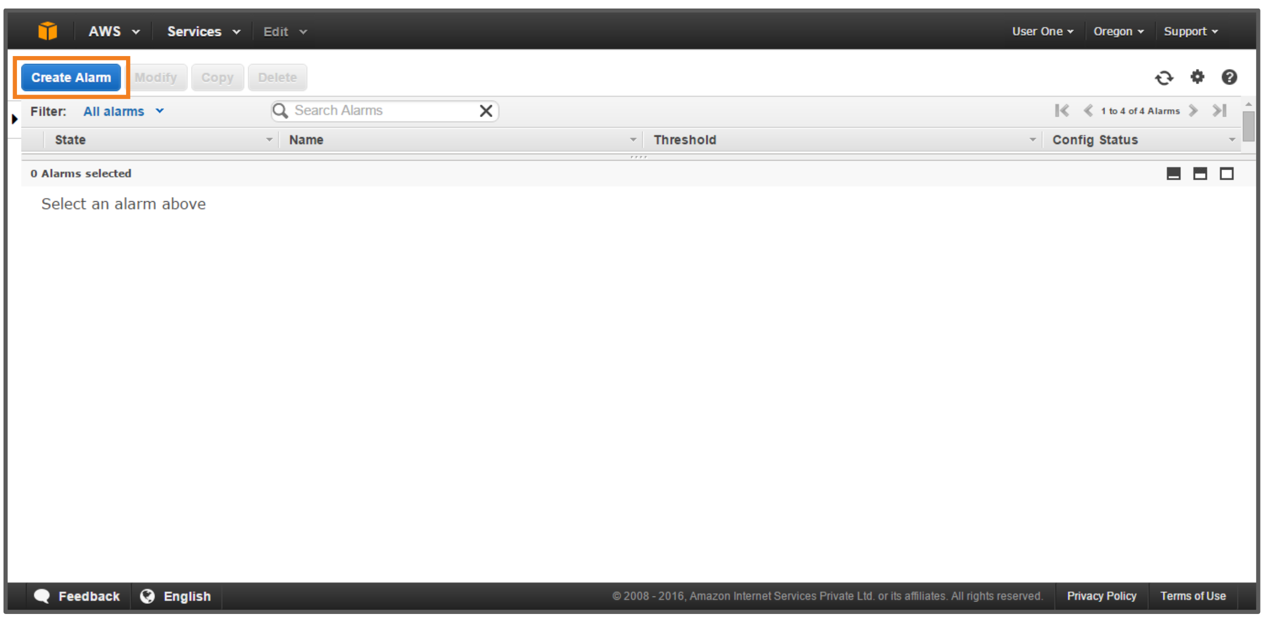
1. Sign in to the AWS Management Console to open the CloudWatch Console.



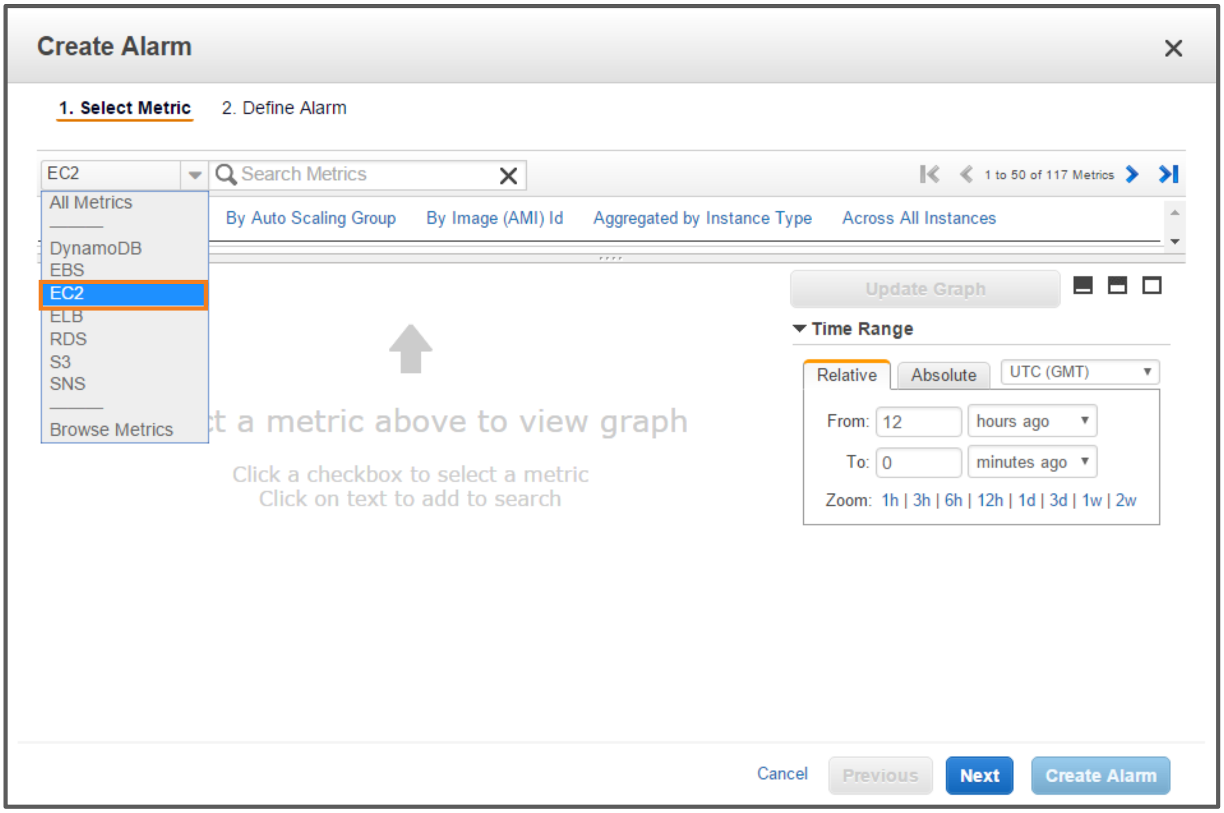
2. On the Navigation pane, click **Alarms**.



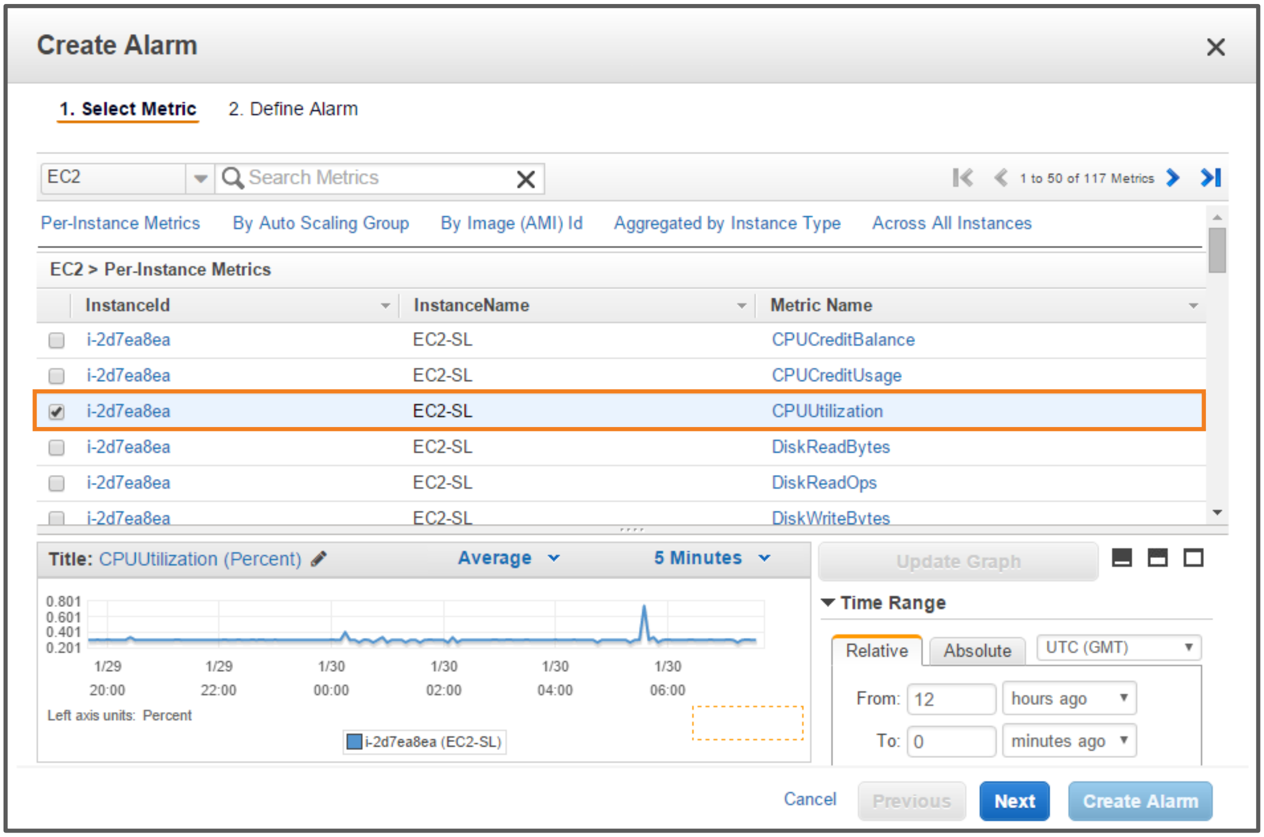
3. Then, click **Create Alarm**.



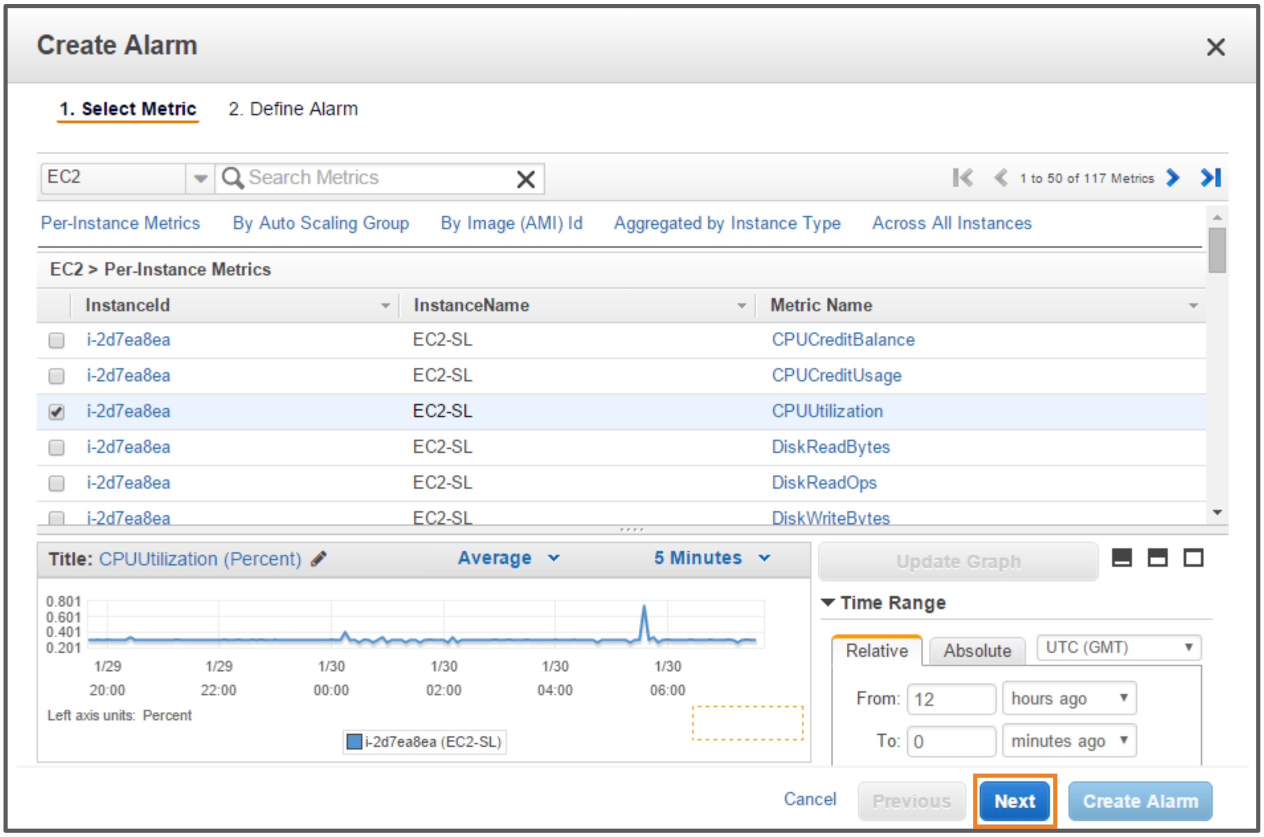
4. In the **Create Alarm** pane, select the **EC2** Metrics category from the Browse Metrics drop down list.



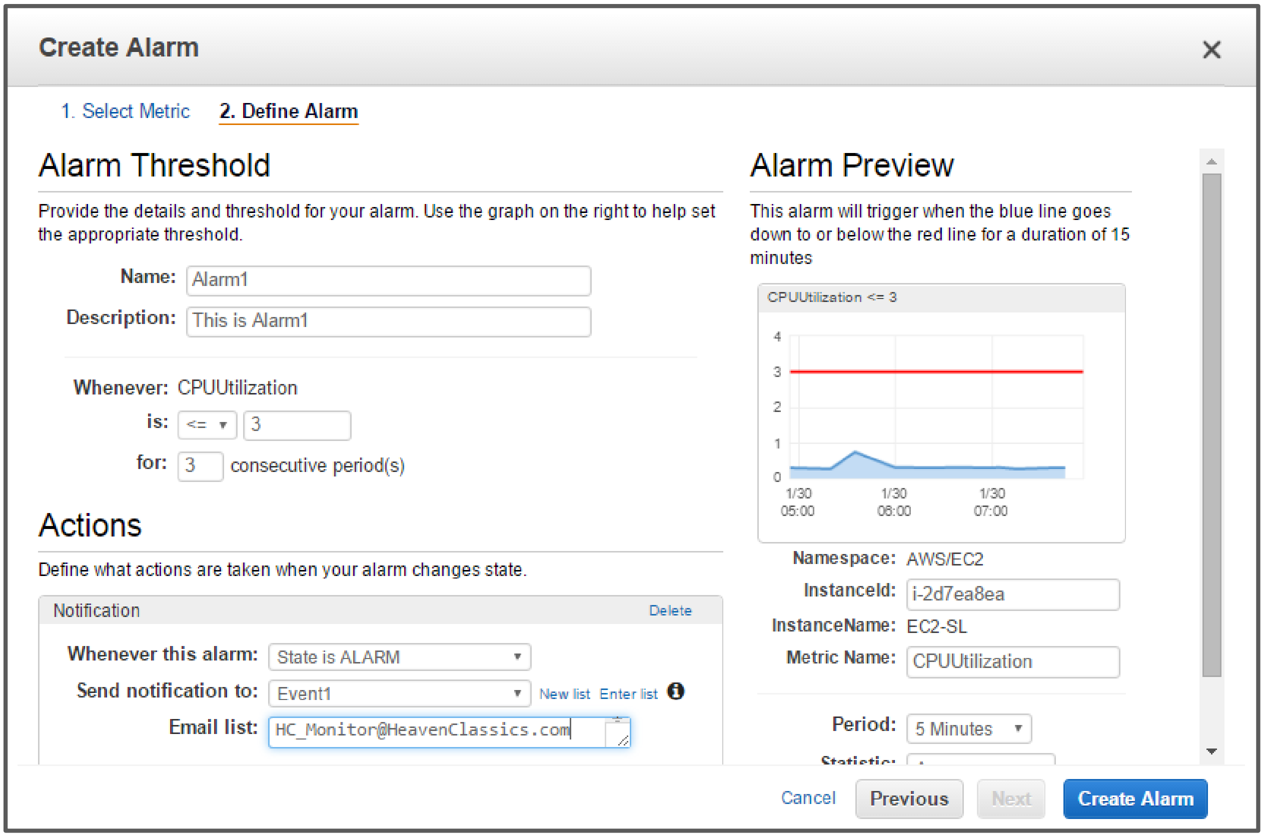
5. Next, select the metric name, CPUUtilization.



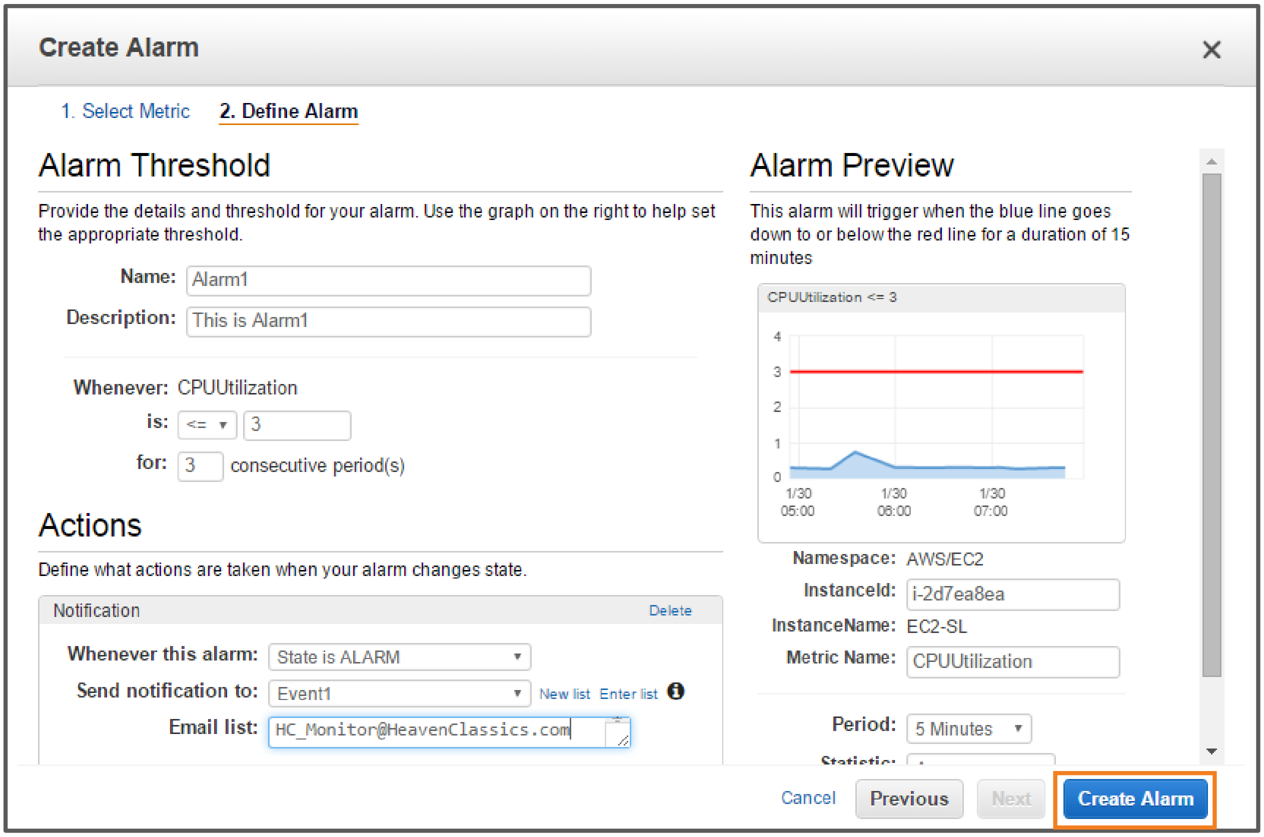
6. Then, click **Next**.



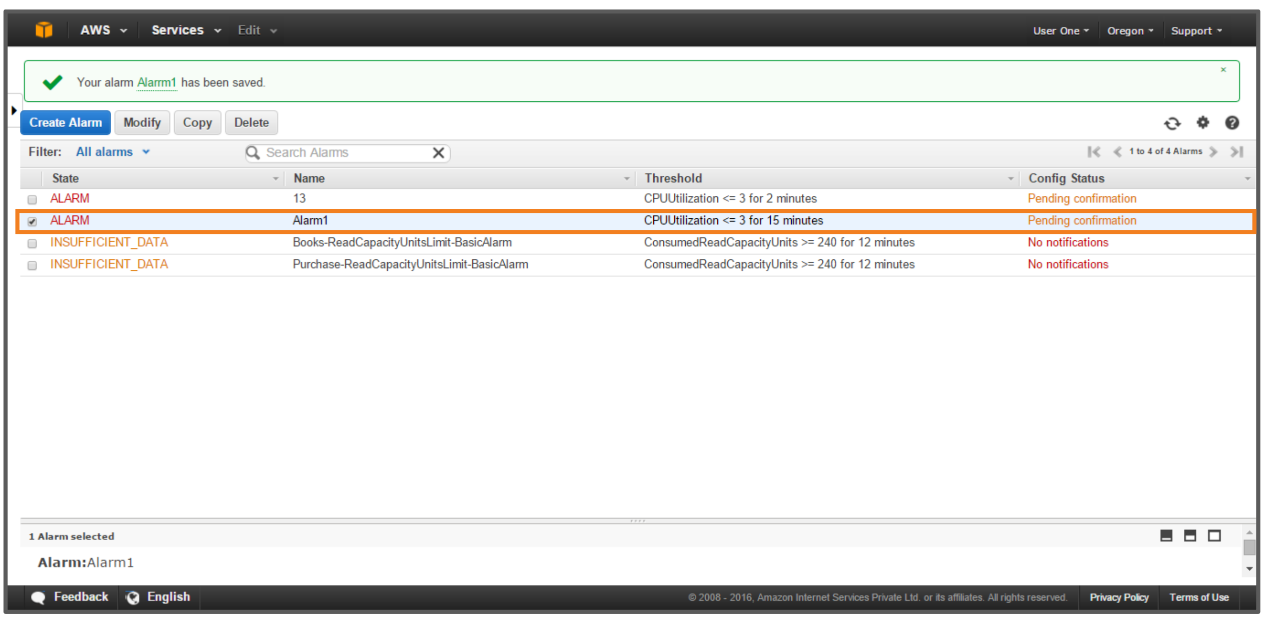
7. In the **Alarm Threshold** section, enter the fields: **Name**: Alarm1, **Description**: This is Alarm1, **is**: <= 3, **for**: 3 consecutive periods(s). In **Actions** section, select the type of action as: **Whenever this alarm**: State is ALARM, **Send notification to**: Event1, **Email list**: HC_Monitor@HeavenClassics.com, **Period**: 5 Minutes.



8. Next, click **Create Alarm.**



9. The Alarm is created.

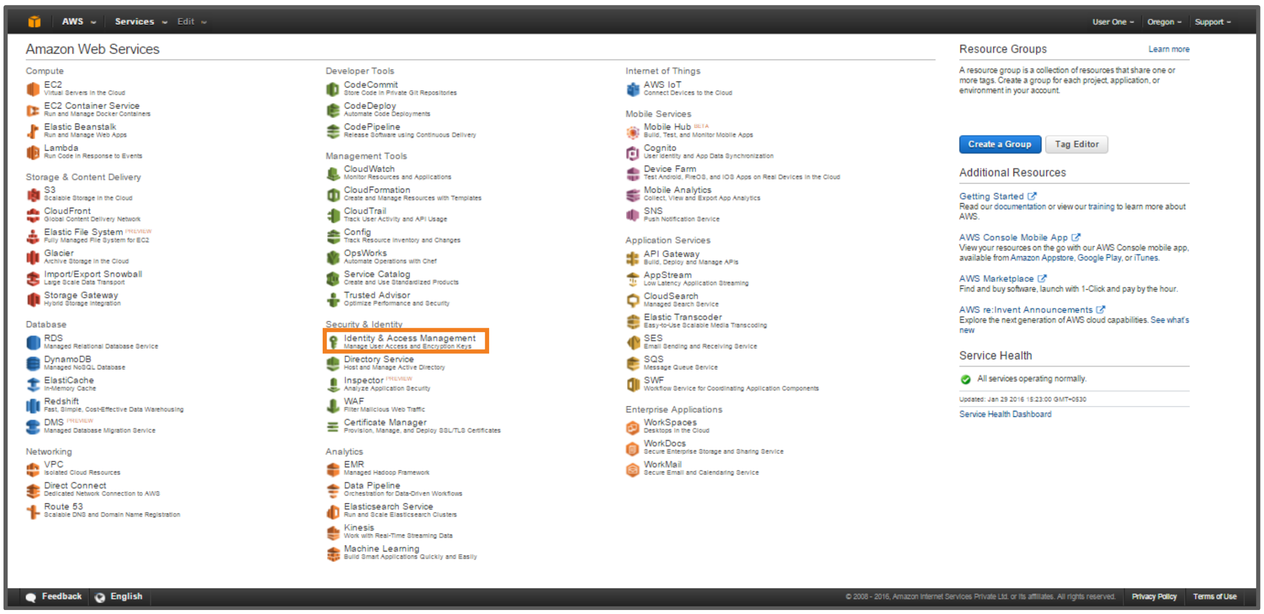


3. *Create an IAM User*

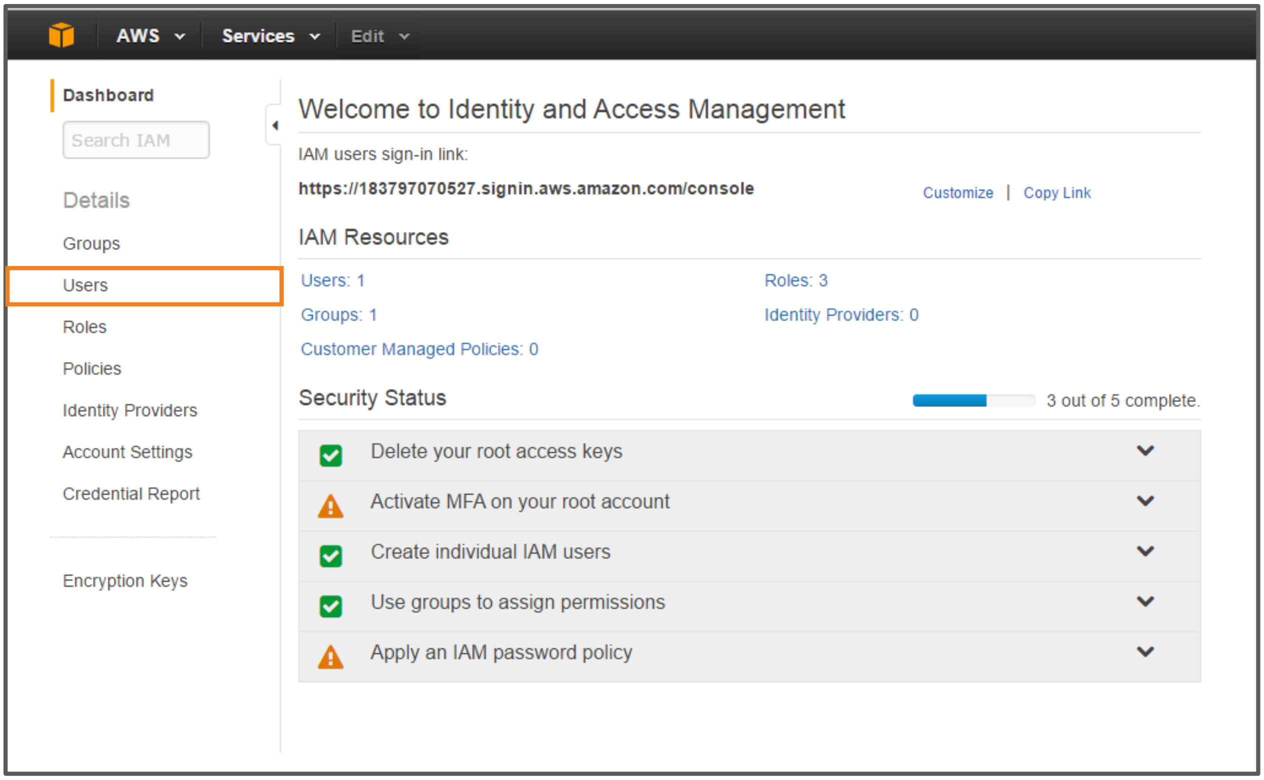
Eventually, the company decided to add employees as Users in the AWS account. Amazon allows to create one or more IAM users in a single AWS account. The support team added existing and new employees as AWS users.

To create an AMI User:

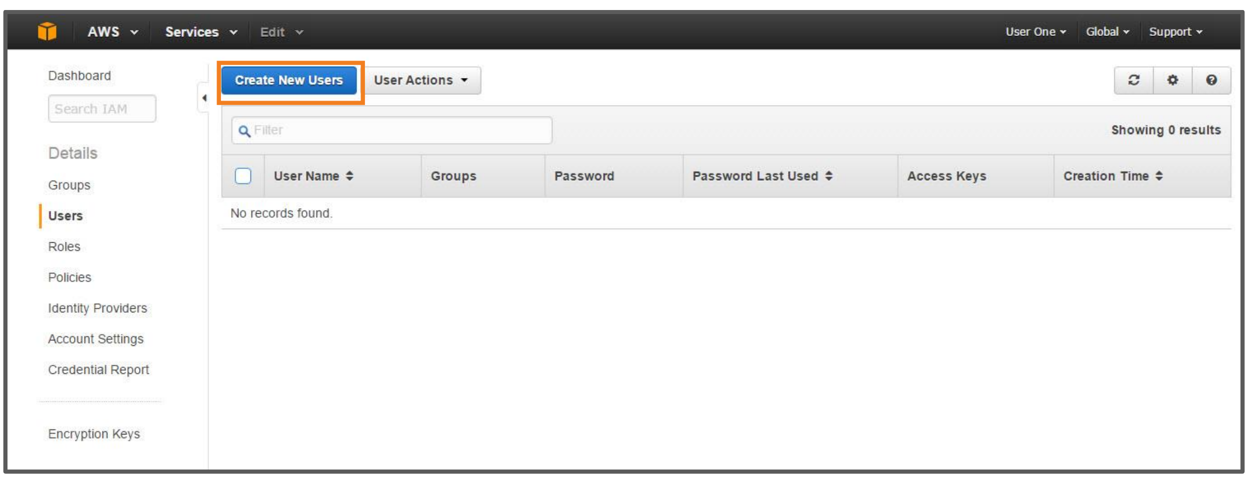
1. Sign in to the AWS Management Console, and open the IAM Console.



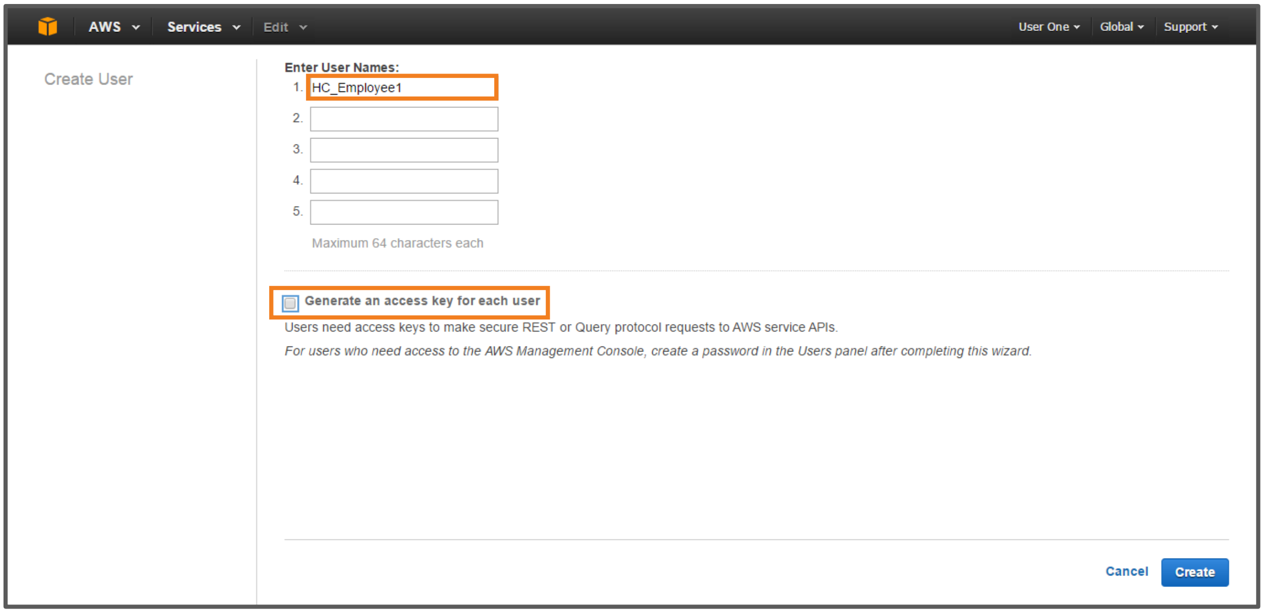
2. On the navigation pane, click **Users**.



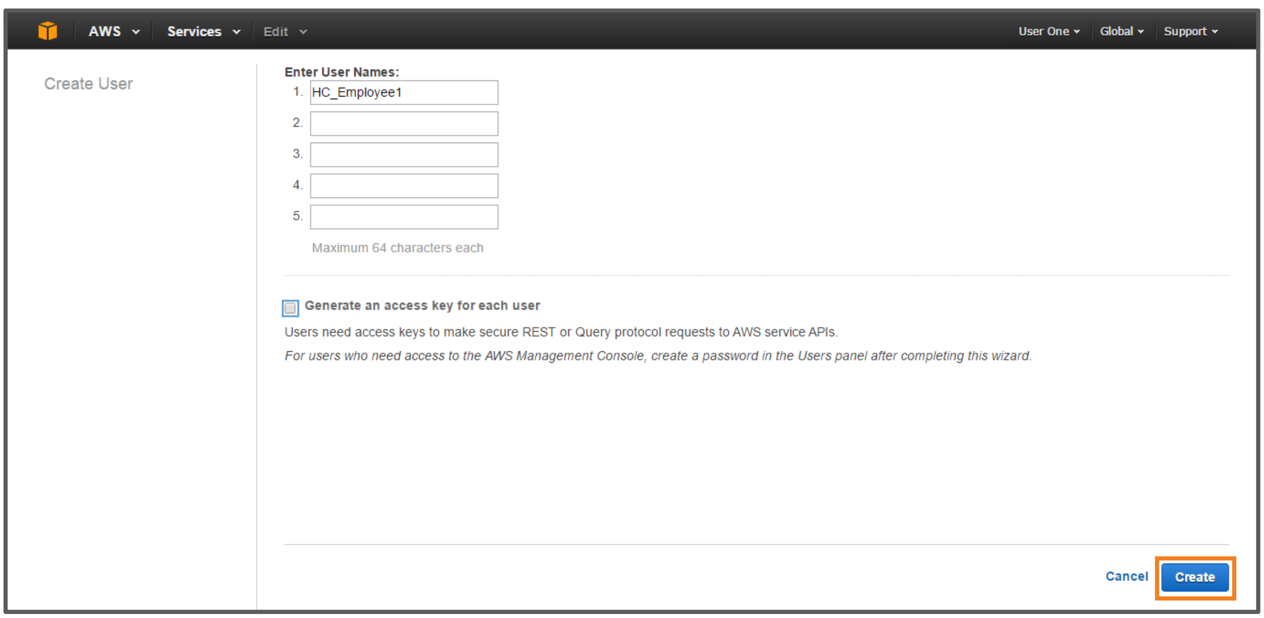
3. Then, click **Create New User**.



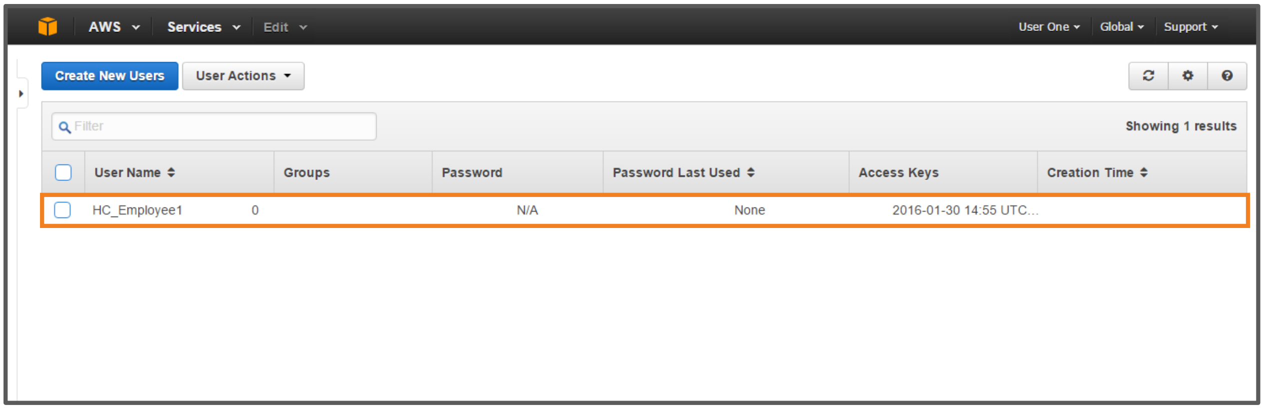
4. In the Create User wizard, in box 1, enter the **User Name** as HC_Employee1. Similarly, you can enter a list of User Names. Next, clear the check box next to **Generate an access key for each user**.



5. Finally, click **Create**.



6. An IAM User is created.

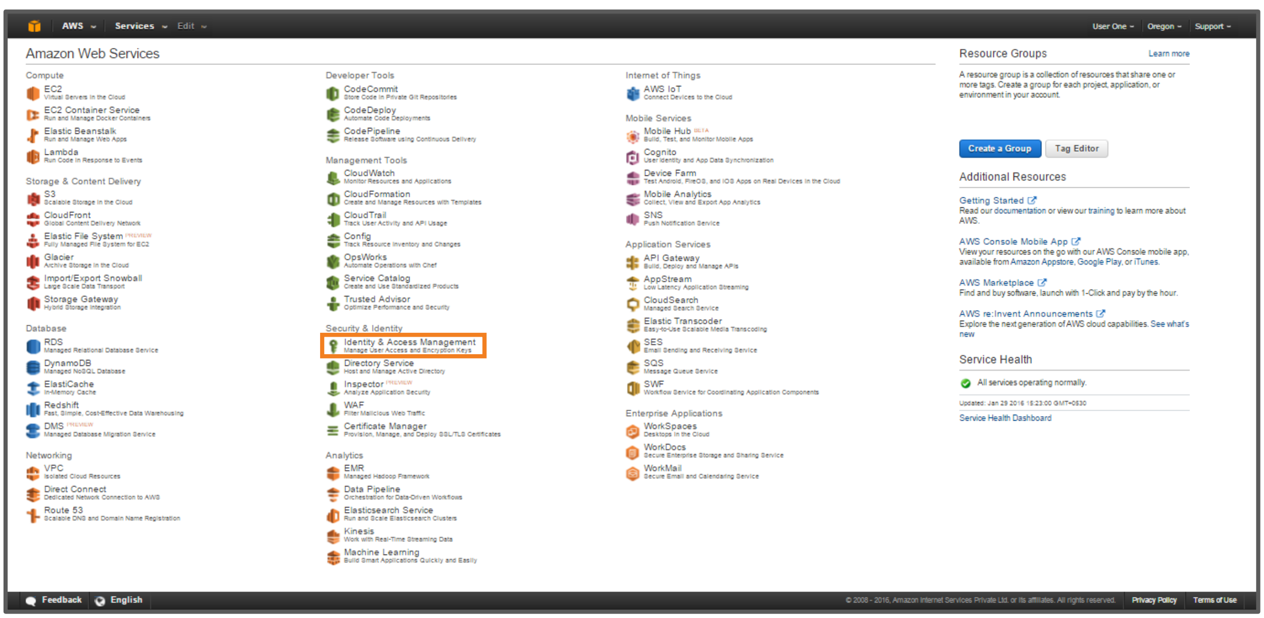


4. Create the IAM Administrator Group, and add the user to the Administrator Group

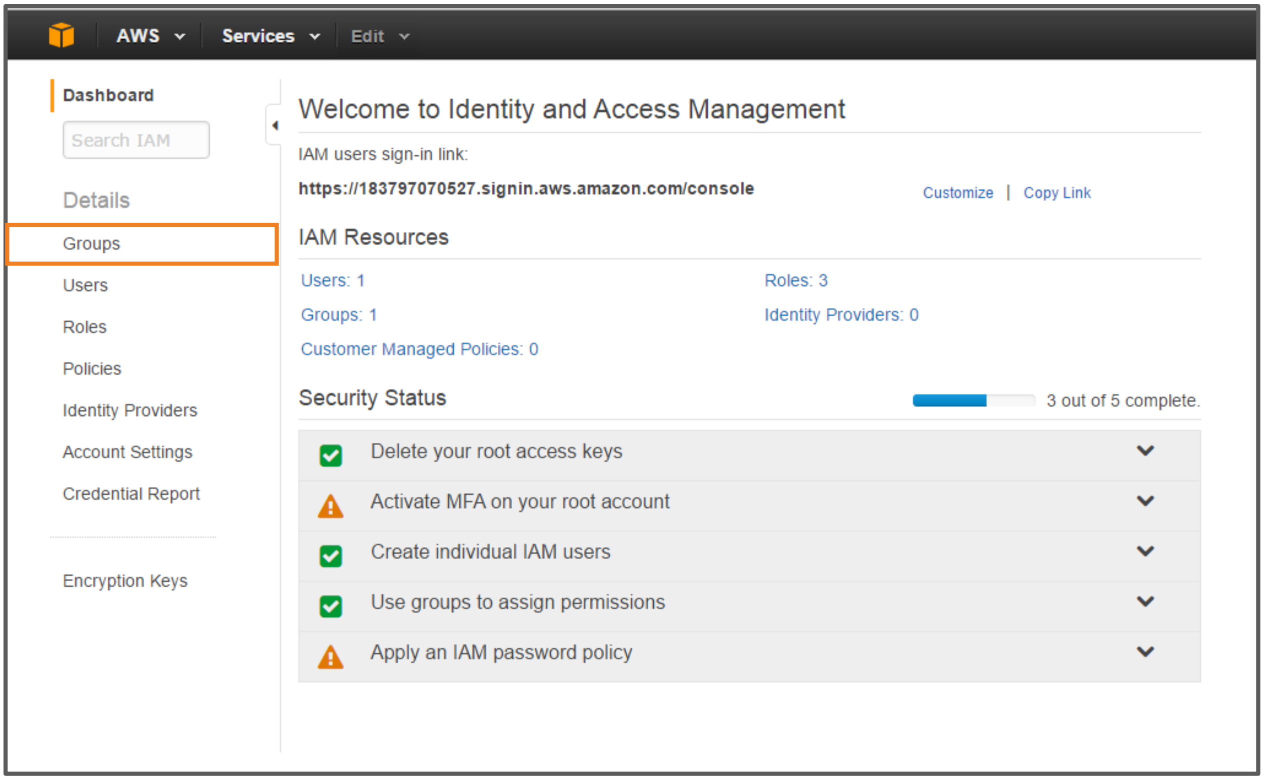
After creating IAM Users, the company formed the IAM Administrator Group. Grouping helped the company to grant a particular permission or allow access for the specified group only.

To create the Administrator Group:

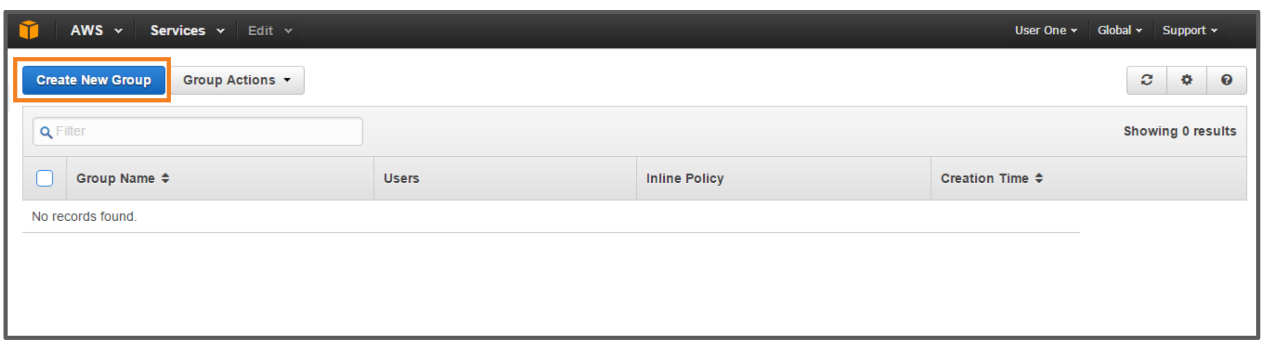
1. Sign in to the AWS Management Console, and open the IAM console.



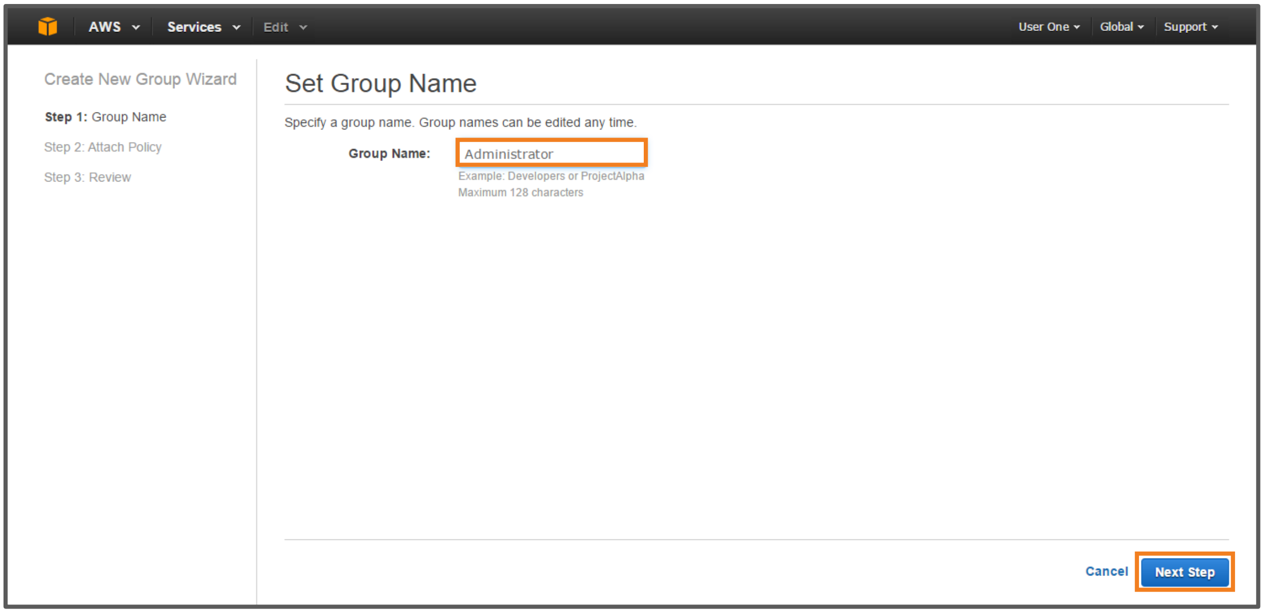
2. On the Navigation pane, click **Groups**.



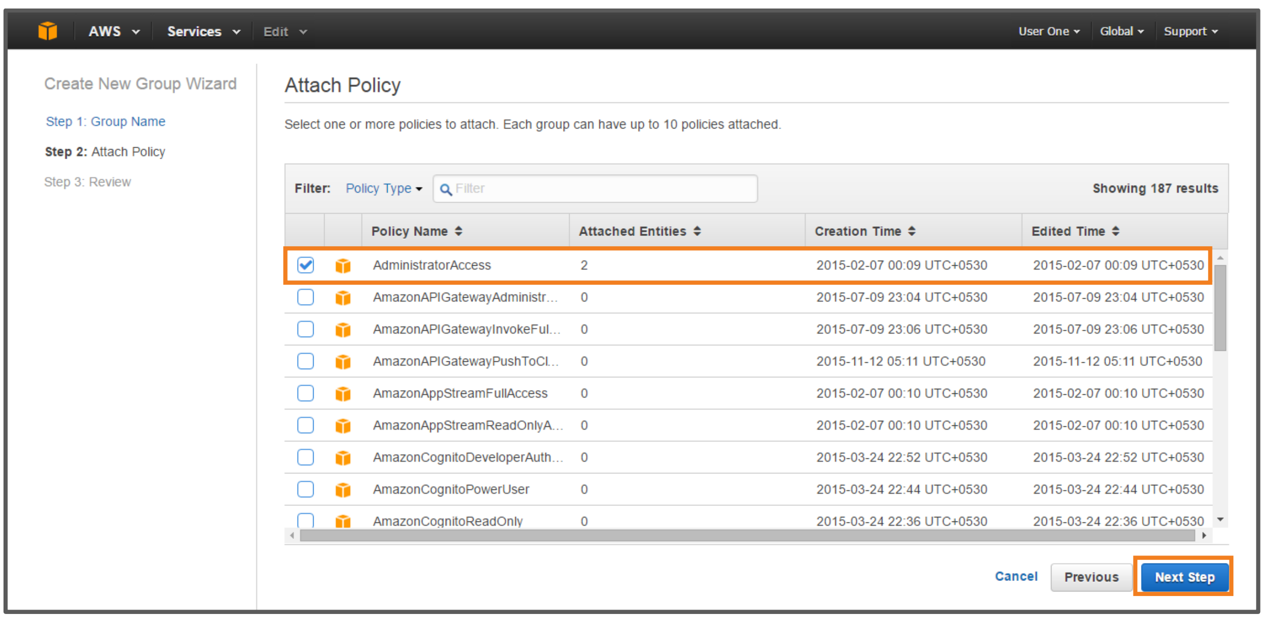
3. Then, click **Create New Group**.



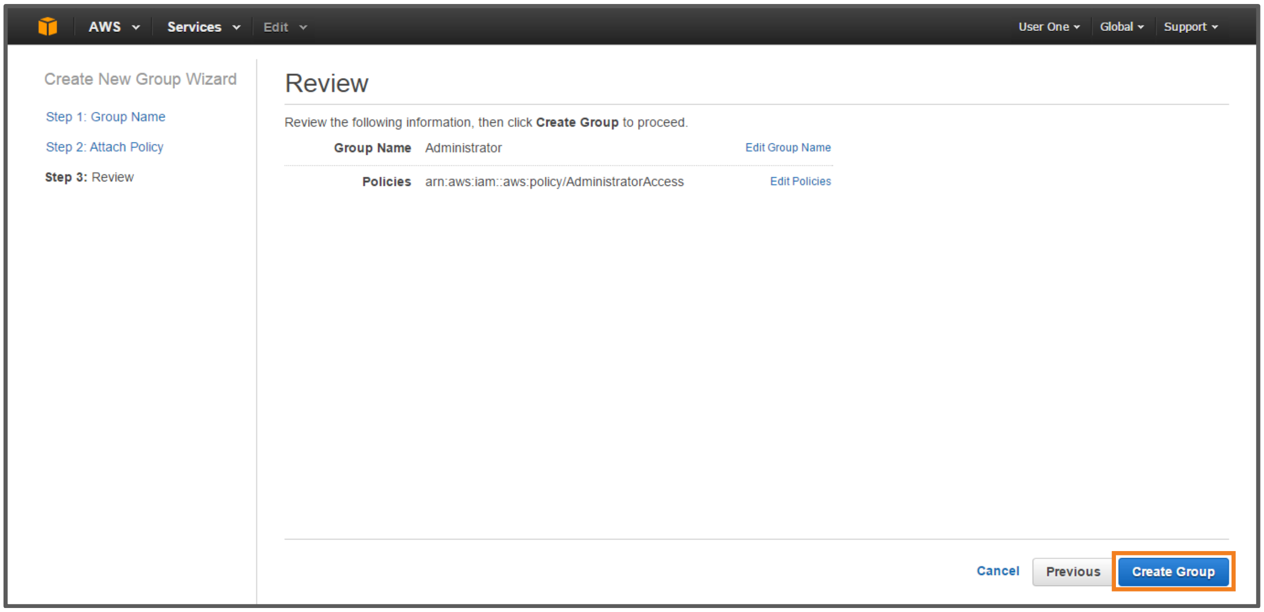
4. In the **Create New Group Wizard**, set the **Group Name** in the given text box as: **Administrator**. Then, click **Next Step.**



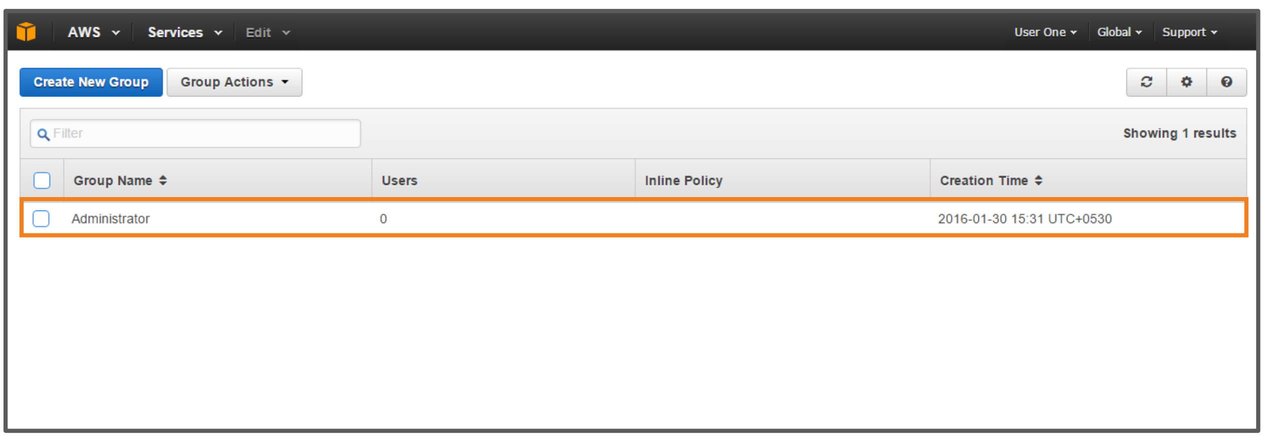
5. Next, assign the policy name as **AdministratorAccess**, and click **Next Step**.



6. Then, review the details of your group request. Finally, click **Create Group**.



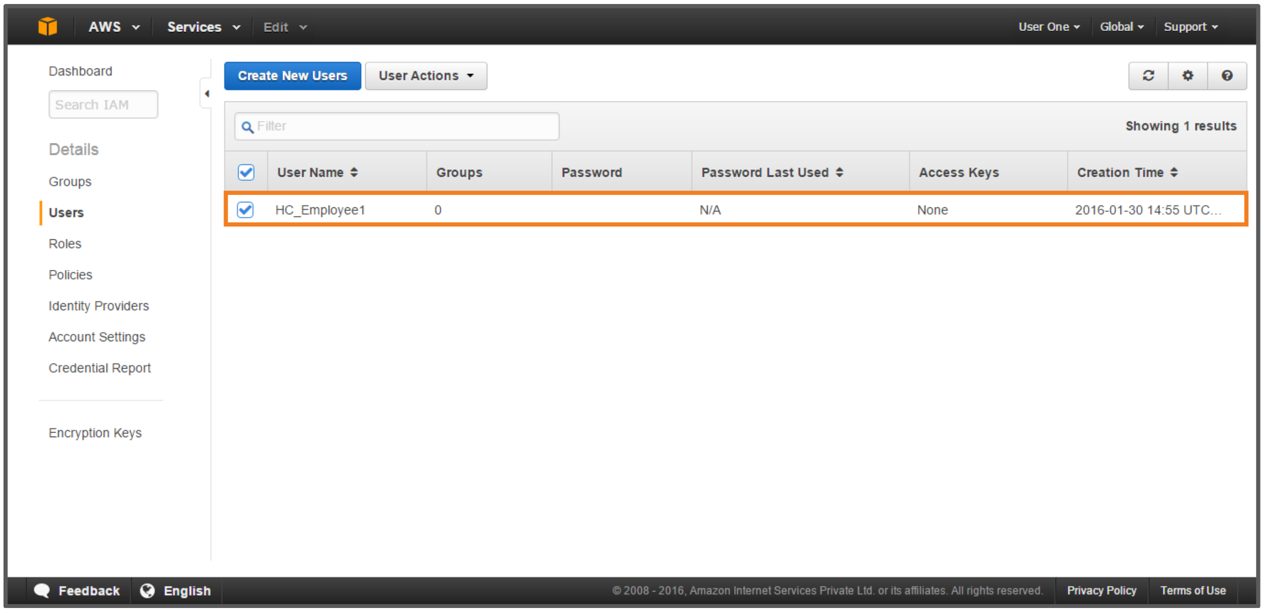
7. A Group is created.



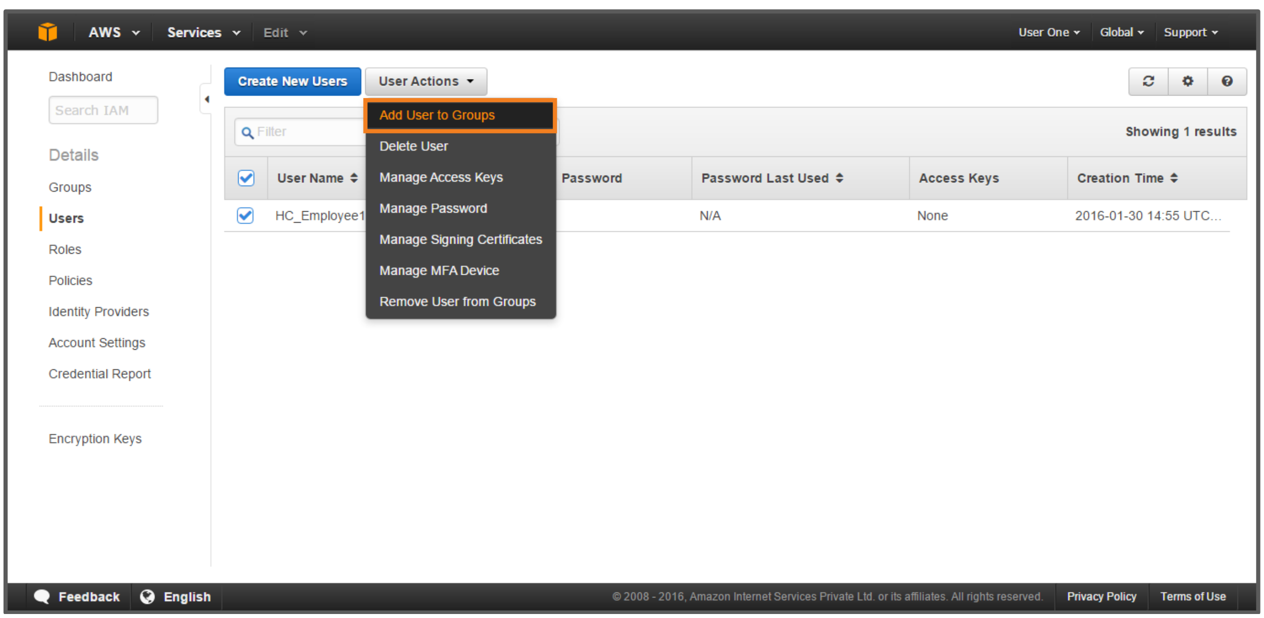
Once the group is created, the team needs to identify ~~the~~ users to be assigned to it. To do this, the team has to add users to the group.

To add Users into the Administrator Group:

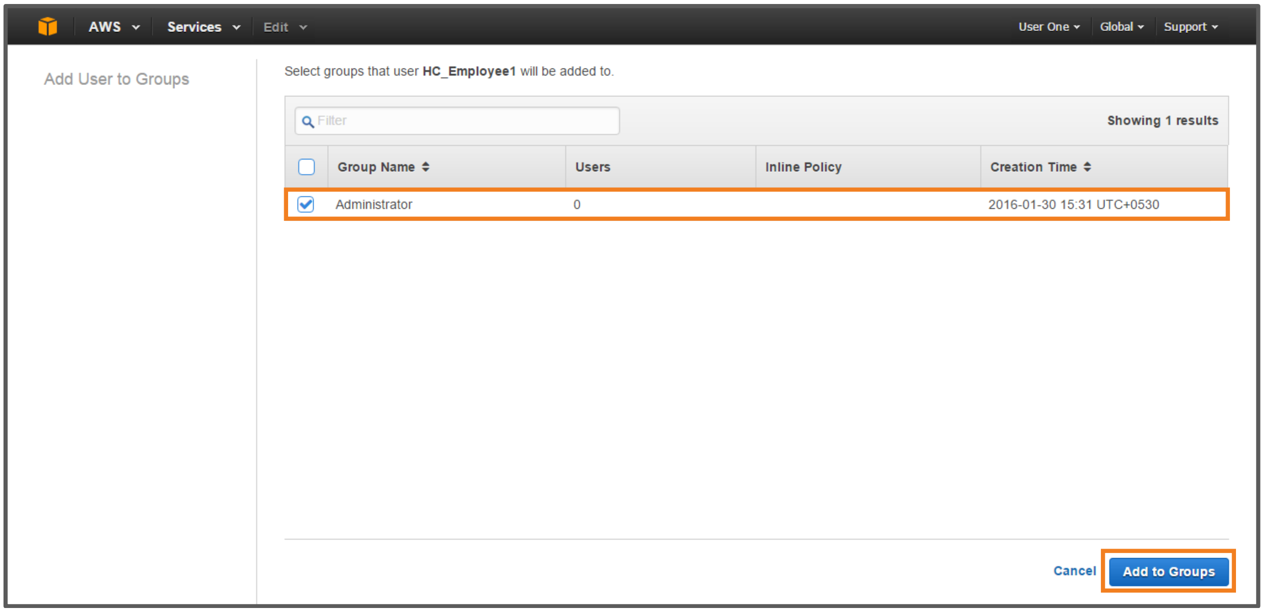
8. From the IAM Console, select the user name to add to the group.
   Select **HC_Employee1**.



9. Click **User Action** drop down, and then click **Add User to Groups**.



10. Here, select the group that needs to be added. Then, click **Add to Groups**.

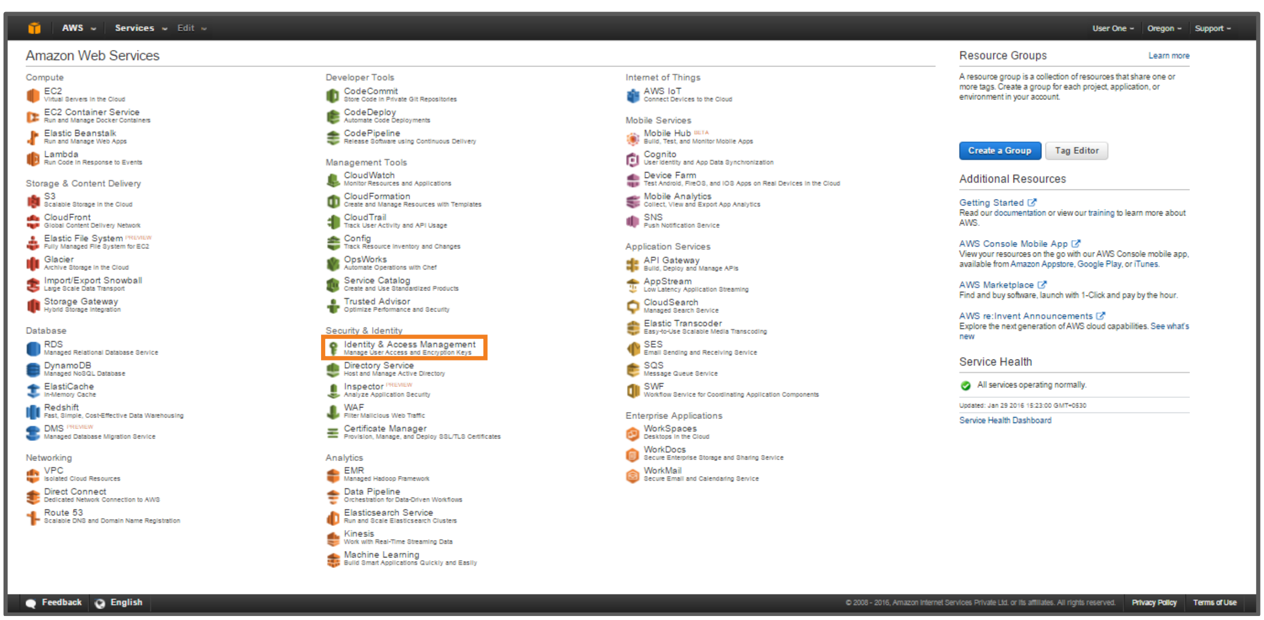


5. Create a Role

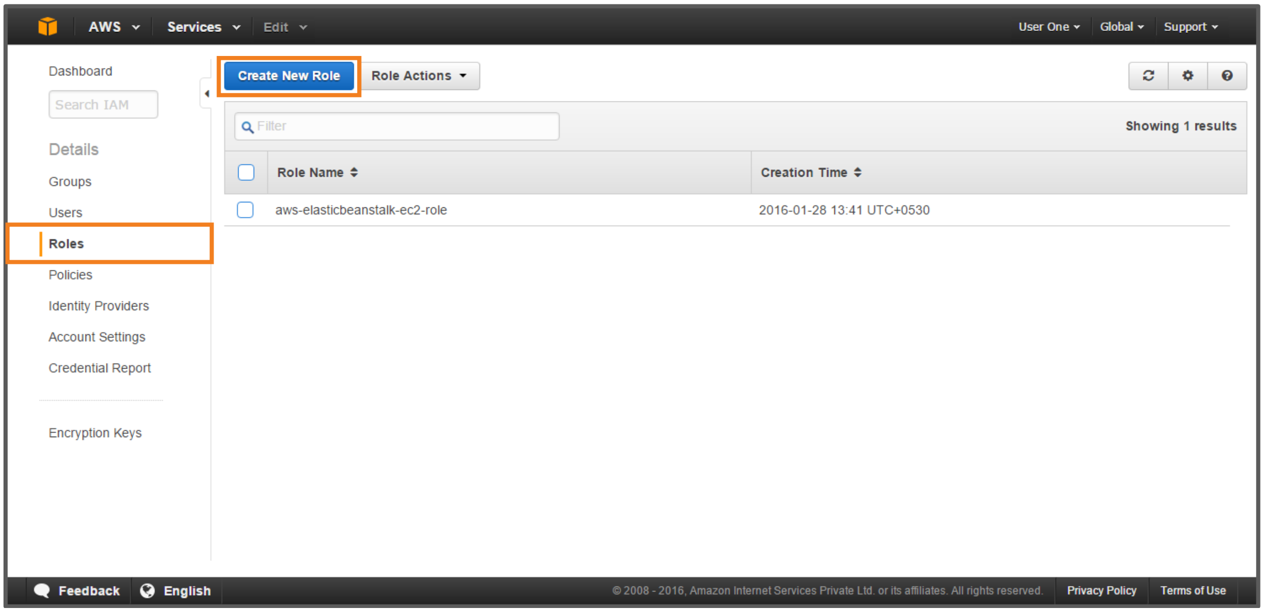
Creating a Role allows access to the AWS resources. This process helps the team to manage user access.

To create a Role:

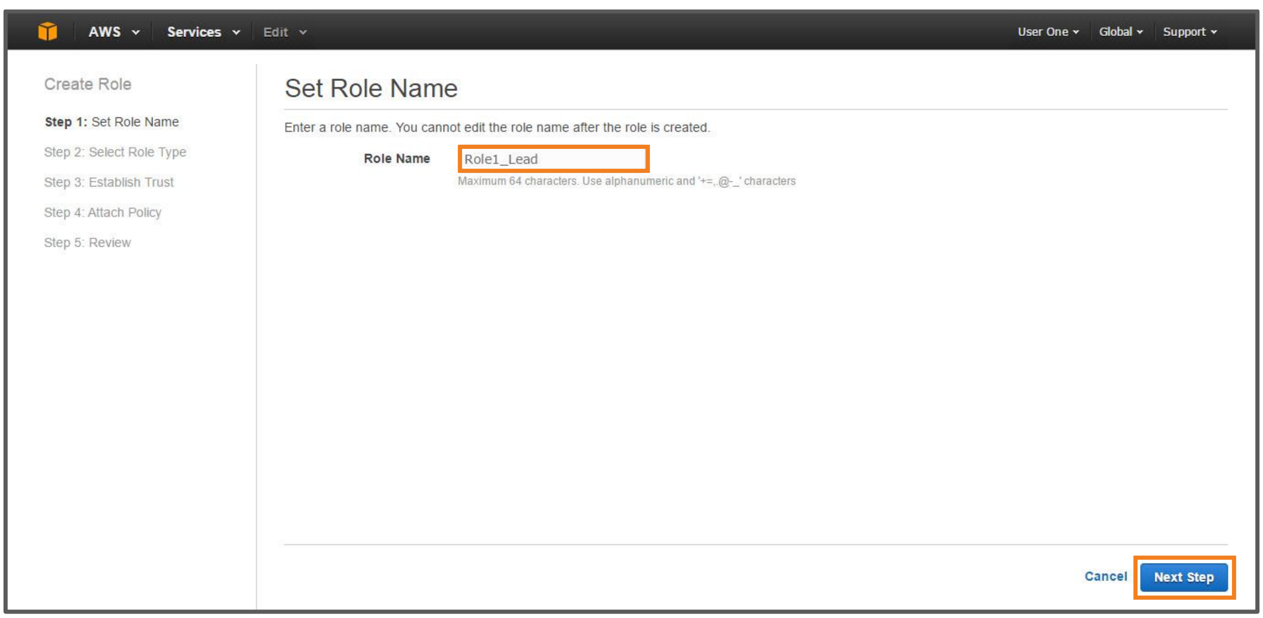
1. Sign in to the AWS Management Console to open the IAM console.



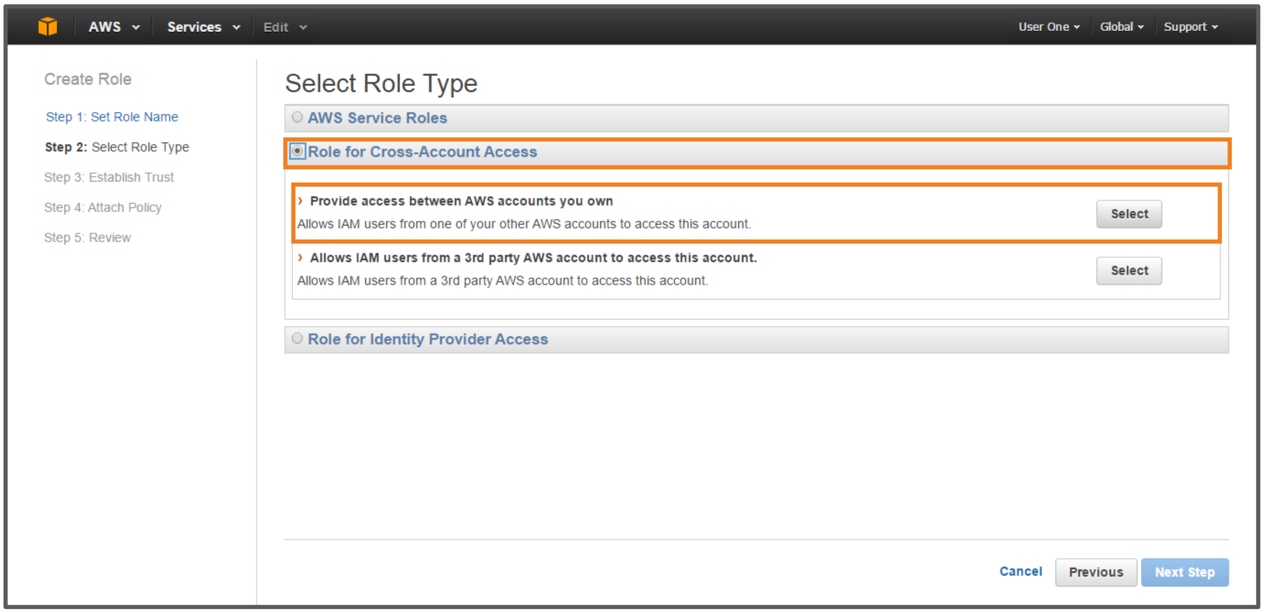
2. On the Navigation pane, click **Roles**. Then, click **Create New Role**.



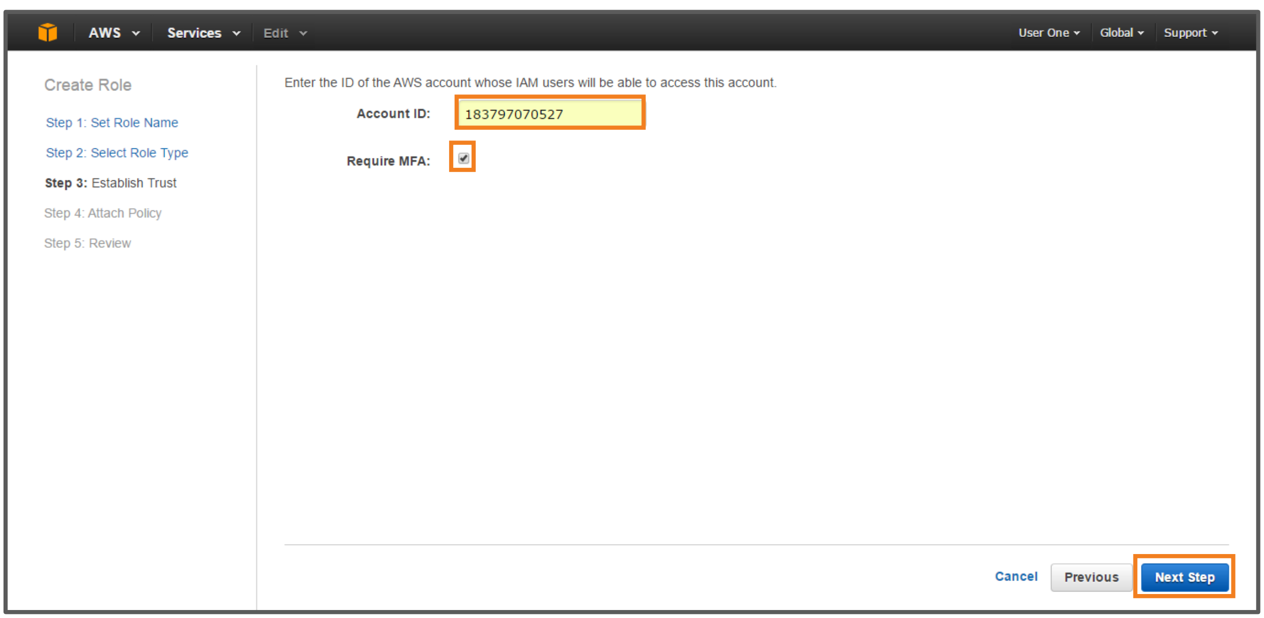
3. In the **Create Role** wizard, type the **Role Name** in the given text box as: **Role1_Lead**. Then, click **Next Step.**



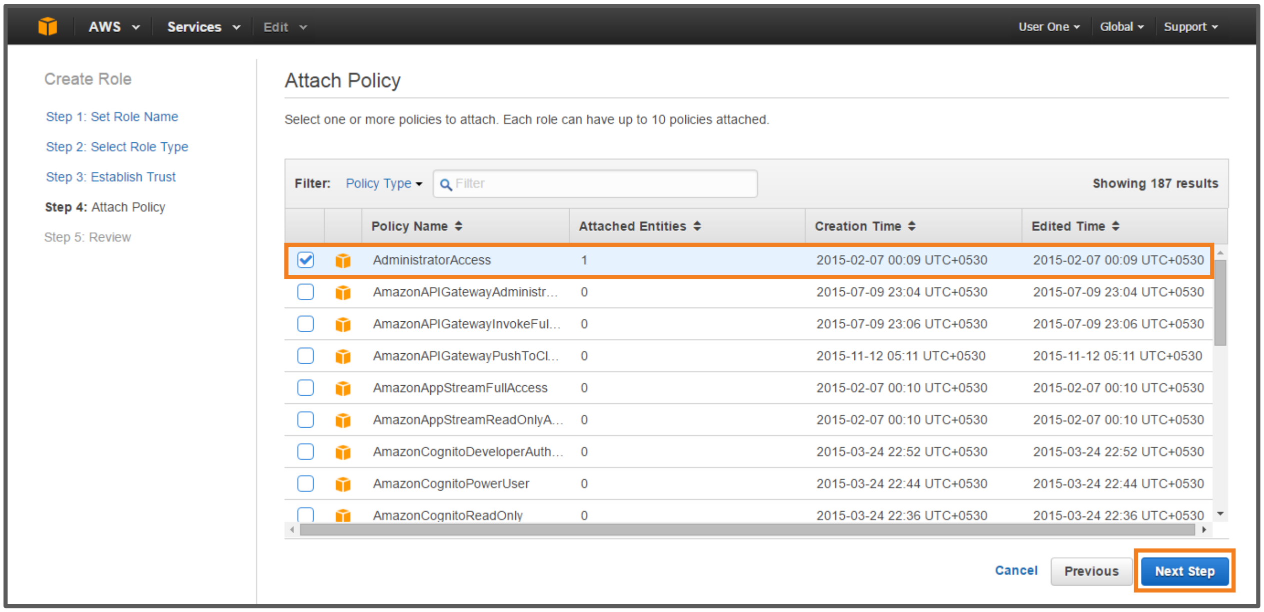
4. The **Select Role Type** page opens. Here, select **Role for Cross-Account Access** section under **AWS Service Roles**, and then select the option as **Provide access between AWS accounts you own**.



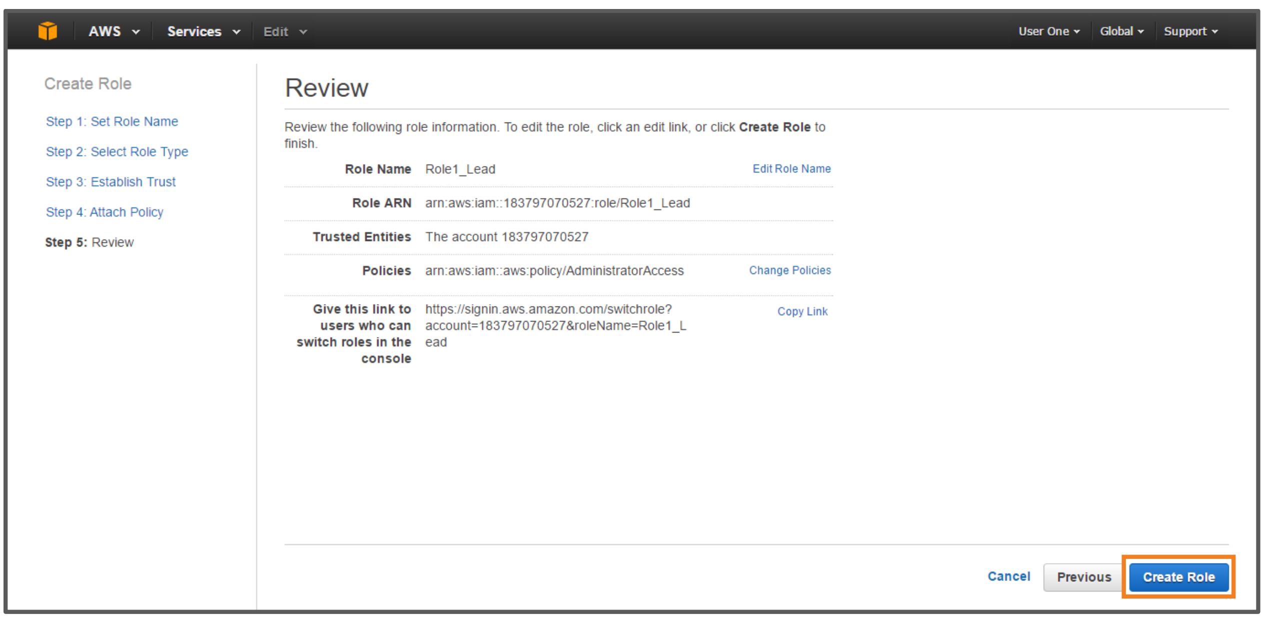
5. Next, specify the AWS **Account ID** to grant access to your resources. Enter the ID of created account: 183797070527. Then, check the **Require MFA** box, and click **Next Step**.



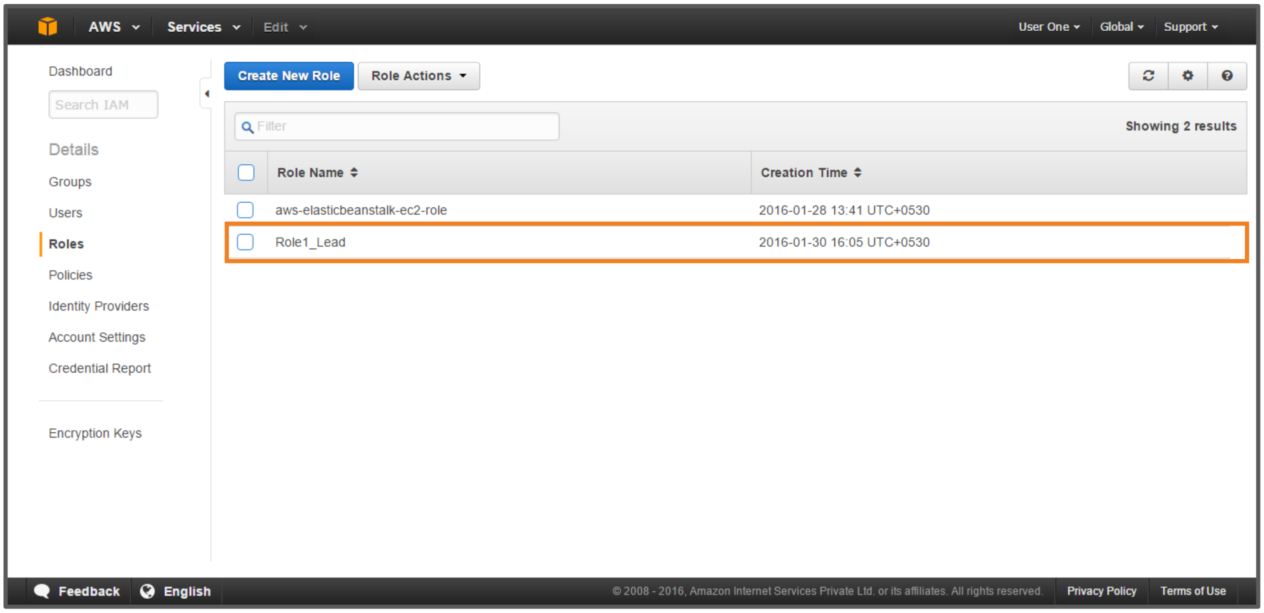
6. On the Attach Policy page, select the **AdministratorAccess** policy. Then, click **Next Step**.



7. On the **Review** page, review the details of your assigned role. Finally, click **Create Role**.



8. A Role is created.


AWS
Services
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Dashboard
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Details
Groups
Users
Roles
Policies
Identity Providers
Account Settings
Credential Report
Encryption Keys
Create New Role
Role Actions
Filter
Showing 2 results
Role Name
Creation Time
aws-elasticbeanstalk-ec2-role
2016-01-28 13:41 UTC+0530
Role1_Lead
2016-01-30 16:05 UTC+0530
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